

A-Visor and A-Camera: Arduino-based Cardboard Head-Mounted Controllers for VR Games

Sangmin Park, Hojun Aan, Junhyeong Jo, Hyeonkyu Kim, Sangsun Han, Jimoon Kim, Pilhyoun Yoon, and Kibum Kim*

Hanyang University, South Korea

ABSTRACT

The Nintendo Labo: VR Kit™ introduced several types of cardboard controllers that allow users to enjoy virtual reality through various interactions. However, it is not compatible with smartphone devices which many people can use to access VR easily. In this study, we used Arduino and a smartphone to create two customized low-cost cardboard head-mounted VR controllers which we called A-Visor and A-Camera. We also created VR games for A-Visor and A-Camera using Unity3D. Thus, we present new DIY head-mounted VR controllers that are made by assembling corrugated cardboard materials, Arduino, and sensors, which are readily accessible to DIY enthusiasts.

Keywords: Cardboard, Virtual Reality, Head-Mounted Device, Arduino, Unity3D.

Index Terms: Human-centered computing-Interaction Paradigms-Virtual reality; Human-centered computing-Interaction devices

1 INTRODUCTION

With advanced VR technology, many people can access virtual reality using a head-mounted device (HMD). HMDs provide visual feedback in virtual reality. Many researchers have tried to add additional modules to HMDs in order to provide realistic feedback [1] and to allow more realistic interaction [2]. The Nintendo Labo: VR Kit™ is a state-of-the-art head-mounted VR controller on HMD in various shapes such as a bird, a camera, a bazooka, and an elephant [3]. Unlike other examples, these controllers are intuitive in that the controllers look the same as their software themes. This kit is a Do-It-Yourself (DIY) VR HMD using cardboard. Many researchers and practitioners developed DIY VR devices like HMD devices [4] or VR controllers [5] starting with Google Cardboard VR. Despite these efforts, the market lacks DIY head-mounted VR input devices that can be applied to smartphone-based cardboard VR. Even though there are similar head-mounted VR controllers such as Nintendo Labo: VR Kit™ [3] and L-Visor [6], these devices are not truly versatile DIY because they are based on the Nintendo Switch device which does not work with other devices like a smartphone.

In response to this lack, we provide the potential for users to develop their own VR game contents and smartphone-based cardboard head-mounted VR controllers. First, we modified the L-Visor [6] cardboard VR controller into the

Arduino as the A-Visor version. Then, we modified a Nintendo Labo: VR Kit™'s camera VR controller and built A-Camera using Arduino in place of the Joy-con of Nintendo Switch. We also developed new applications for A-Visor and A-Camera controllers using Unity3D. Through this process, we proposed and developed a low-cost, head-mounted VR input device in a smartphone-based cardboard VR system. In this paper, we show an example of an intuitive and interactive HMD for smartphone-based cardboard VR.

2 HARDWARE

To develop a small size module that fits into the controller, we used Arduino. We built the Arduino sensor module using the following modules and sensors. We used the Arduino Nano Every™ board because it is small and capable of computing general code. We used the accelerometer/gyroscope module (MPU-6050) which detects the visor's tilting angle in the A-Visor controller and camera lens rotation angle in the A-Camera controller. Our devices take data from the user's tilting and rotating interactions. We also used the Bluetooth module (HC-06) which consists of an RX pin as the data receiver, TX pin as data the sender, plus power and GND pin. The module takes the input of the sensor and sends the data to a paired smartphone.

To make A-Camera and A-Visor, we used L-Visor [6] and Nintendo Labo: VR Kit™'s camera controller respectively. The Arduino sensor module was attached on the left side of the A-Visor to detect the angle of the visor's tilt and another was attached in the middle of the A-Camera to detect the rotation angle of the camera lens. Thus, the Arduino sensor module can track the user's tilting interaction with A-Visor



Figure 1: A-Visor interaction



Figure 2: A-Camera interaction

* email address: kikum@hanyang.ac.kr

and the user's rotating interaction with A-Camera. Both controllers allow users to interact in VR easily and intuitively. To assemble controllers and cardboard VR HMD, we made an additional foldable adapter for each controller. The assembled controller, adapter, and cardboard VR HMD and its controller are shown in Figures 1 and 2.

3 SOFTWARE

To develop Bluetooth communication between Arduino and Unity, we used a smartphone Bluetooth serial communication plugin (ArduinoBluetoothAPI). Using API, we added sending and reading functions to the connected devices. For reading, we utilized ReceiveManager which recognizes the input data that Arduino sends. ReceiveManager saves the serialized data from the accelerometer/gyroscope module. SendManager sends the data to the Unity3D application. In Unity, the system gets the serialized data repeatedly using the Update method to keep track of the user's interaction. From the input data, Unity uses a low pass filtered data of accelerometer and gyroscope module to track the controller's movement. Through this code, the application shows the changing environment caused by the user's interaction.

We developed two games for each controller. First, a transparent game allows the player to tilt A-Visor repeatedly. This game aims to find all the red balls inside buildings and objects such as boxes. When the player tilts the visor controller down once, the objects of the left area (indicated as "1" in the left figure of figure 3) become transparent using a diffuse shader. And then, the next building to the right (indicated as "2" in the left figure of figure 3) becomes transparent when the player tilts the visor controller down twice. Each time the player tilts the visor down, the assigned object will be translucent in the order shown on the left of figure 3. When the player finds a red ball, the player can erase that ball by staring at it with the Raycast code.



Figure 3: Transparent Game
(Left: Overall view at the start of the game,
Right: A view that the No.4 object is transparent)

Second, a focus game allows the user to turn the A-camera lens right and left repeatedly. We expect this game will give the user the experience of focusing on the image. The focus game starts at the left of figure 4. This game aims to focus on five different defocused views. When the game begins, a blur filter attached to the camera object applies the blur effect with a random value of interpolation in the range from -1.5 to 1.5. When the player tilts the camera lens controller, the value interpolation is also changed. When the value of interpolation reaches the range from -0.05 to 0.05, the blur effect almost disappears. When the player has focused on all five images, the game ends.

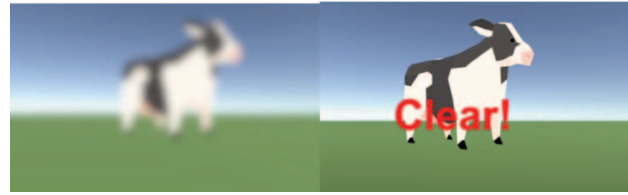


Figure 4: Focus Game
(Left: A view at the start of the game,
Right: A view when in focus)

4 DISCUSSION AND CONCLUSION

We developed DIY examples of a head-mounted controller for a smartphone-based cardboard head-mounted VR controller which consists of Arduino based sensors and modules. This study developed and demonstrated potentials for various interactions with DIY cardboard VR controllers, a rotating camera lens, and a tilting visor. However, the hardware can be improved with an expansion board or other communication methods like WiFi. As a future work, we plan to develop other DIY cardboard controllers like Nintendo Labo: VR Kit™ 's Bird and Lever. A-Visor and A-Camera show an example of a new, cost-effective interactive head-mounted controller for smartphone-based cardboard VR.

ACKNOWLEDGEMENT

This research was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF), funded by the Ministry of Education (No. 2018R1D1A1A02085645).

REFERENCES

- [1] R. L. Peiris, W. Peng, Z. Chen, L. Chan, and K. Minamizawa, "ThermoVR: Exploring Integrated Thermal Haptic Feedback with Head Mounted Displays," in *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 2017: Association for Computing Machinery, pp. 5452–5456, doi: 10.1145/3025453.3025824. [Online]. Available: <https://doi.org/10.1145/3025453.3025824>
- [2] R.-C. Chang, C.-H. Chiang, S.-w. Hsu, C.-Y. Yang, D.-Y. Huang, and B.-Y. Chen, "TanGo: Exploring Expressive Tangible Interactions on Head-Mounted Displays," in *Symposium on Spatial User Interaction*, 2020: Association for Computing Machinery, p. Article 3, doi: 10.1145/3385959.3418457. [Online]. Available: <https://doi.org/10.1145/3385959.3418457>
- [3] Nintendo. "Nintendo labo." <https://labo.nintendo.com/> (accessed 10.30, 2020).
- [4] Maxime-coutte. "Relativ - Build Your Own VR Headset for \$100." <https://create.arduino.cc/projecthub/relativty/relativ-build-your-own-vr-headset-for-100-57adba> (accessed 01.19, 2021).
- [5] S. Yu, Y. Hsu, G. Chen, and J. Peng, "Virtual reality survival first person shooting game," in *2017 International Automatic Control Conference (CACs)*, 12-15 Nov. 2017 2017, pp. 1-5, doi: 10.1109/CACS.2017.8284241.
- [6] S. Park et al., "L-Visor: Visor Input Device for Labo Head-Mounted Display," presented at the Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems, Honolulu, HI, USA, 2020. [Online]. Available: <https://doi.org/10.1145/3334480.3383052>.